

Wildfire Containment Squad

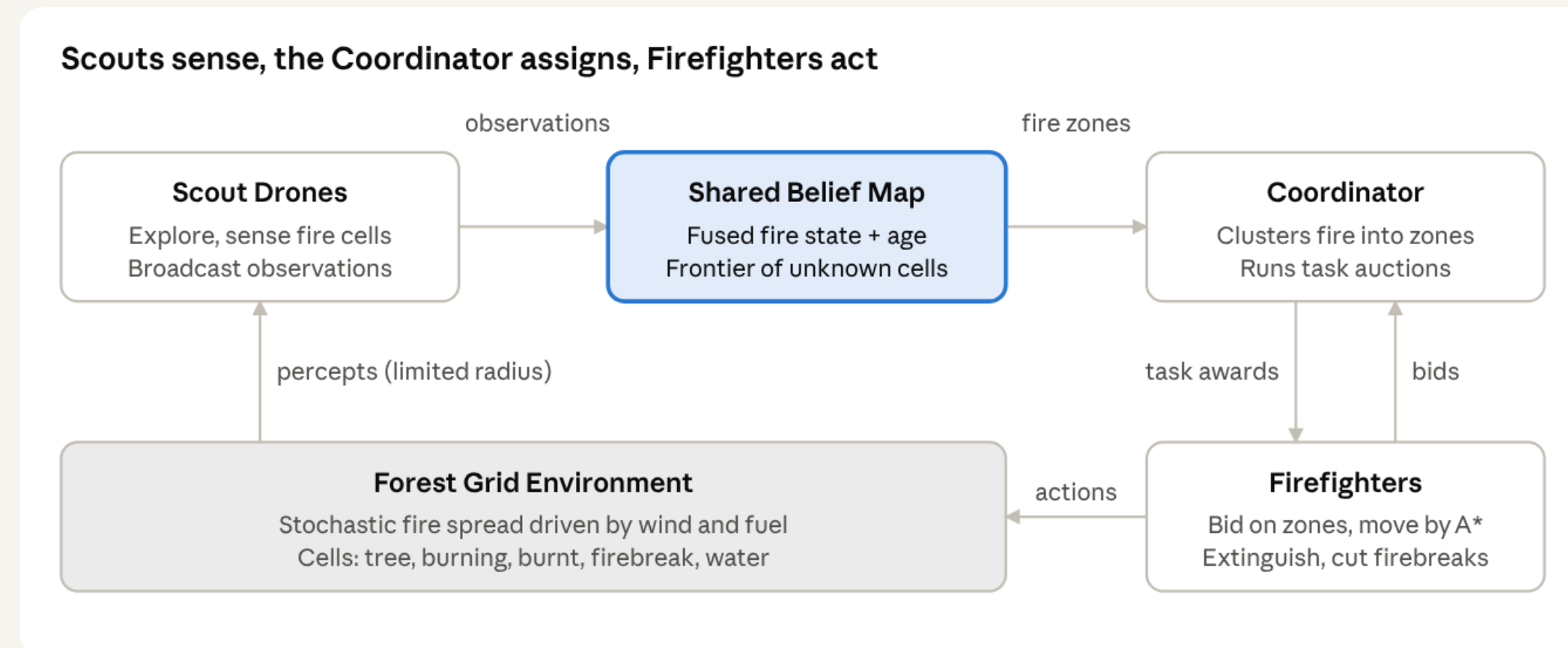
Scout drones, firefighters and a coordinator containing a wildfire together: built, demonstrated and tested.

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RECAP OF THE REVIEW 1 DESIGN

No single agent can see or reach the whole fire



3 scout drones see 3 cells around them and map the fire.

1 Coordinator has no body; it clusters fire into zones and auctions them.

5 firefighters see 1 cell, bid, travel with A* and fight.

Each tool was chosen for one job

Tool	Used for	Why this one
Mesa 3.5	Agents, grid, data collection	Built for agent-based models; seeded randomness included
Solara	The live browser demo	Mesa's own UI stack; play, step, reset, sliders
NumPy	Fire spread, risk maps	Whole-grid spread in 8 array shifts: 1.4 s per run
heapq	A* and UCS queues	Standard library, no extra dependency
PyYAML	config.yaml	Every parameter in one readable file
pytest, pandas, Matplotlib	Tests, tables, charts	114 tests; reproducible CSVs and PNGs

One-click setup (**setup.bat** / **setup.sh**), pinned versions, and a clean install verified on Python 3.12 and 3.13.

Six fixed stages per step, seven messages

Every step, in this order

- 1. Scouts sense, send OBSERVE, fly
- 2. Coordinator clusters and auctions (every 5 steps, or when a firefighter is freed)
- 3. Firefighters act in id order, reserving cells
- 4. Fire spreads (every 5th step)
- 5. Anyone on a burning cell is caught
- 6. Metrics; stop if no fire or step 300

One message bus

Message	From → to
OBSERVE	Scout, firefighter → map
TARGET	Scout → other scouts
ANNOUNCE	Coordinator → free firefighters
BID	Firefighter → Coordinator
AWARD	Coordinator → winner
DONE	Firefighter → Coordinator
REVOKE	Coordinator → firefighter

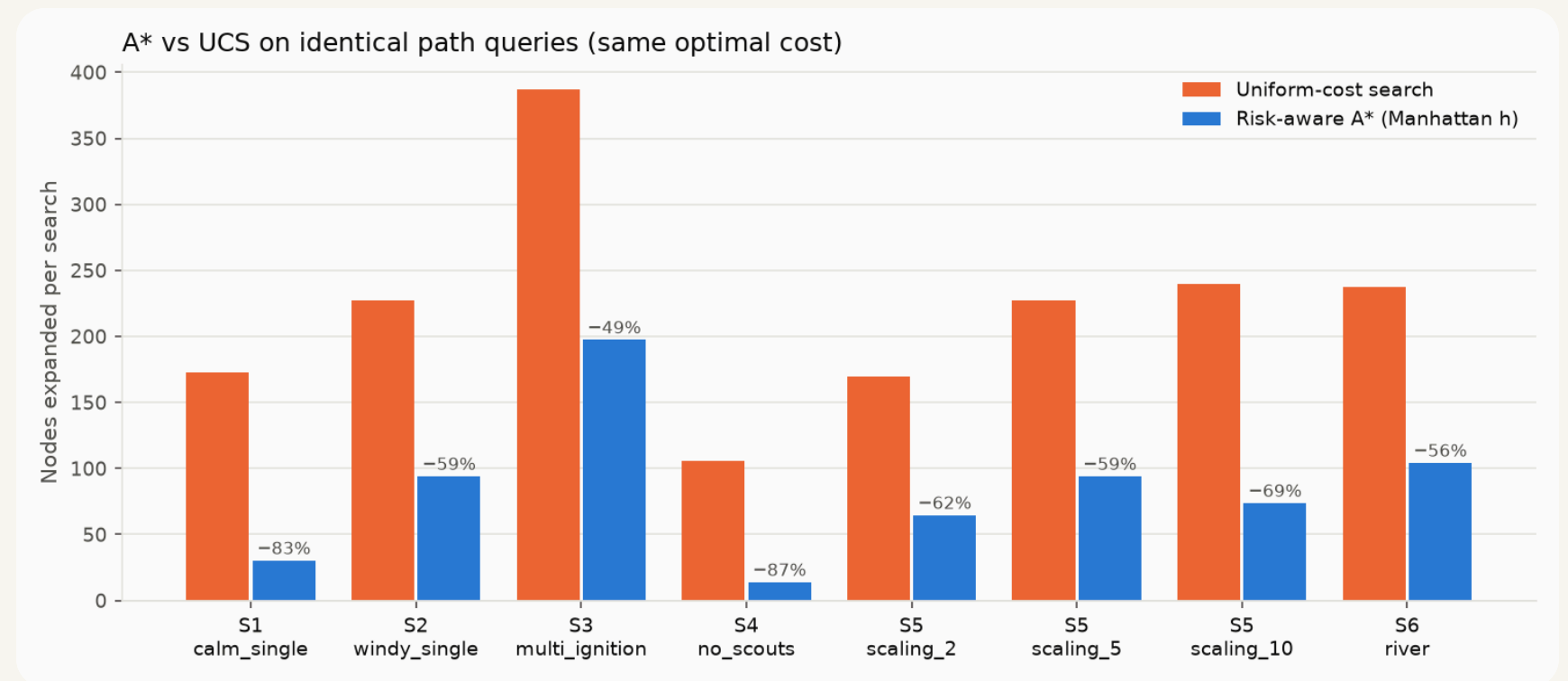
Risk-aware A* and a threat-first auction

Navigation: risk-aware A*

Step cost = $1 + 5 \times \text{risk of the cell entered}$. Manhattan heuristic: admissible and consistent, so paths are optimal.

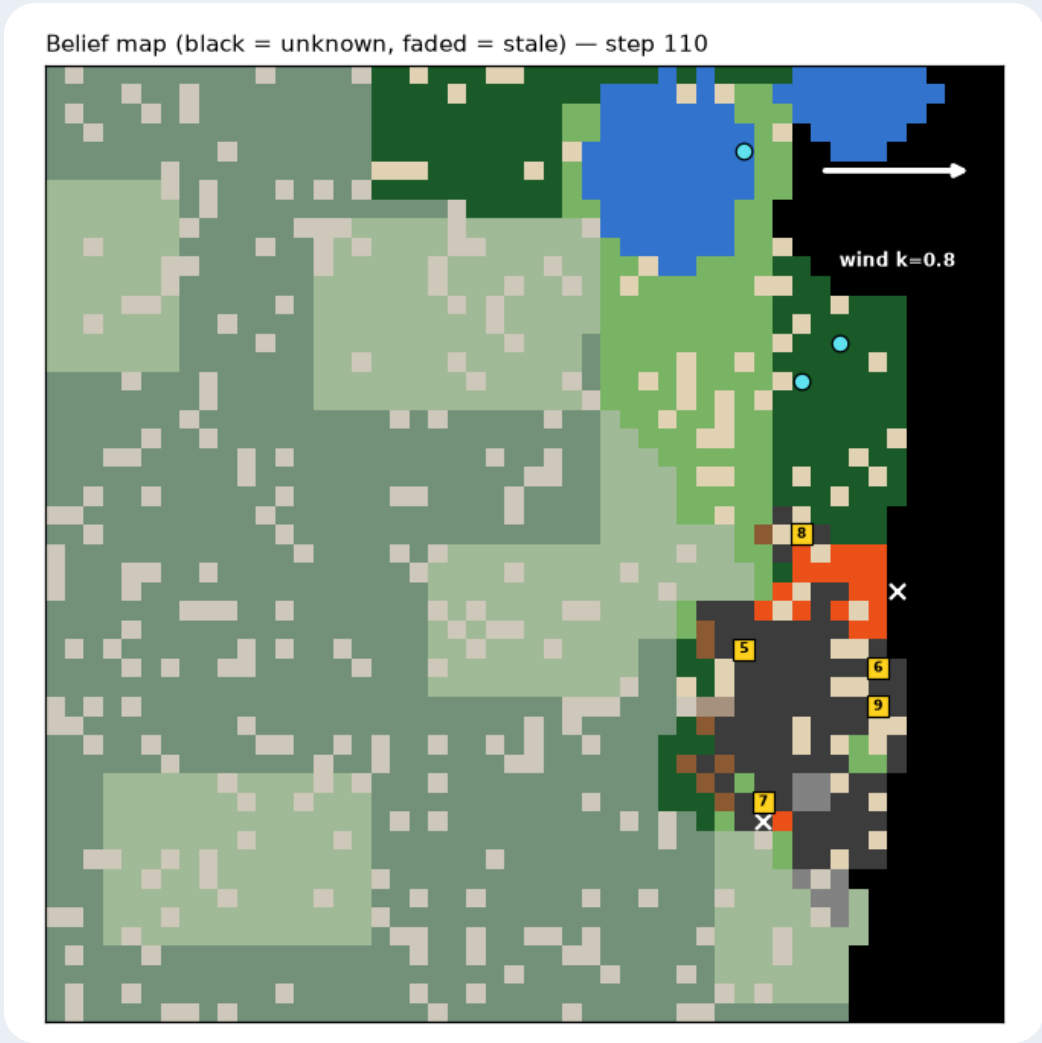
Allocation: sequential auction

Bid $U = \text{threat} \div (1 + \text{path cost}) \times \text{water} \div 10$. Highest threat first; 2 slots above 20 cells.



LIVE DEMO

Same fire, three strategies



What the team knows: black = unknown, faded = stale



One click runs all three strategies on the identical fire

TESTING SCENARIOS

Three layers of testing on identical fires

114

automated tests, 89% coverage:
every rule, conflict and formula

720

experiment runs: 6 scenarios × 3
strategies × 30 seeds

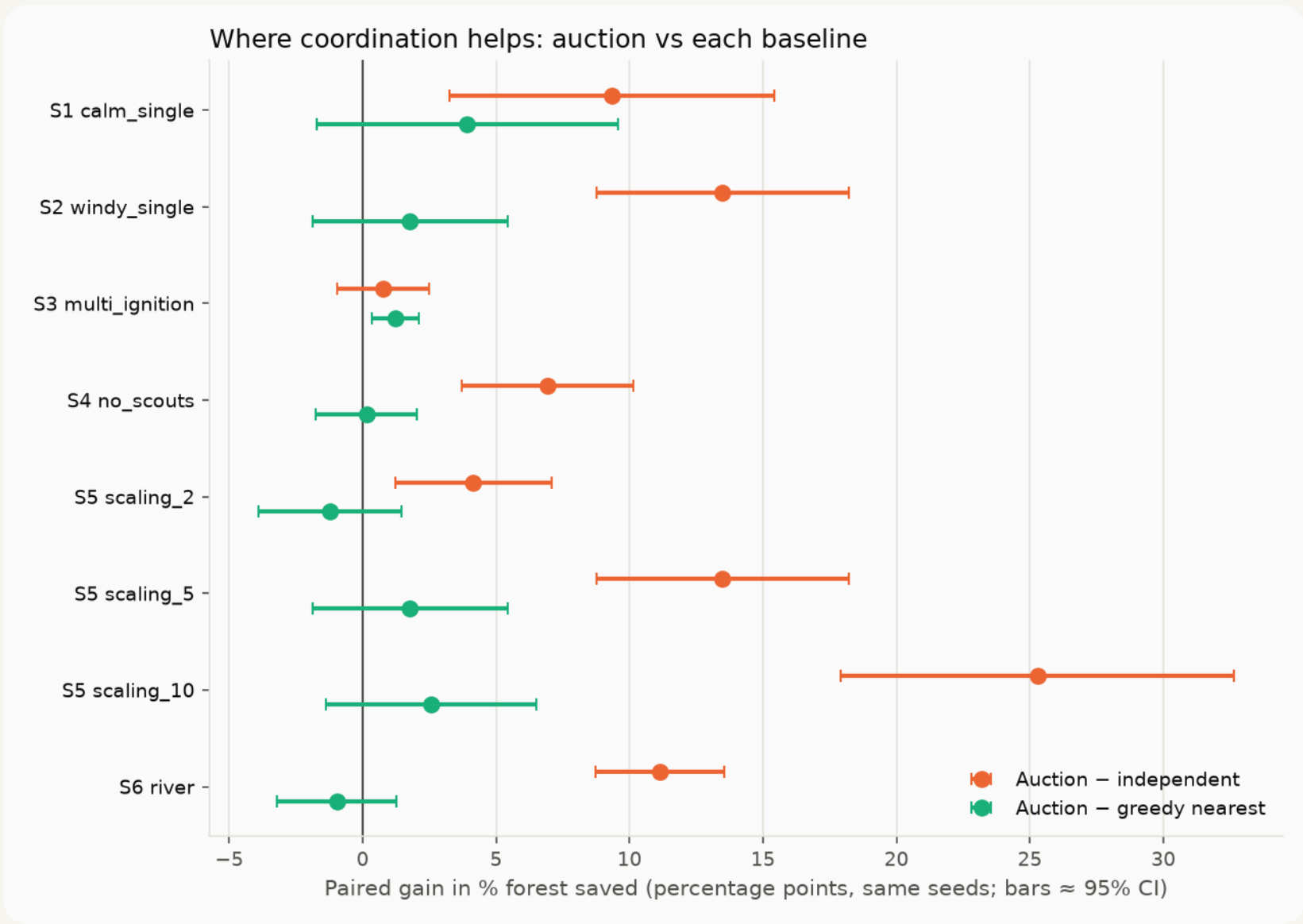
1,090

study runs testing each design
decision

Scenarios: S1 calm · S2 strong wind · S3 three fires · S4 no scouts · S5 2, 5 and 10 firefighters · S6 river with two crossings

Paired design: a seed gives every strategy the same map and the same fire, so we report the per-seed difference with a 95% confidence interval.

Coordination wins in 7 of 8 configurations



+25.3 points

more forest saved with 10 firefighters; +13.5 with 5

29 of 30

river fires handled better than by independent agents

+1.2 ± 0.9

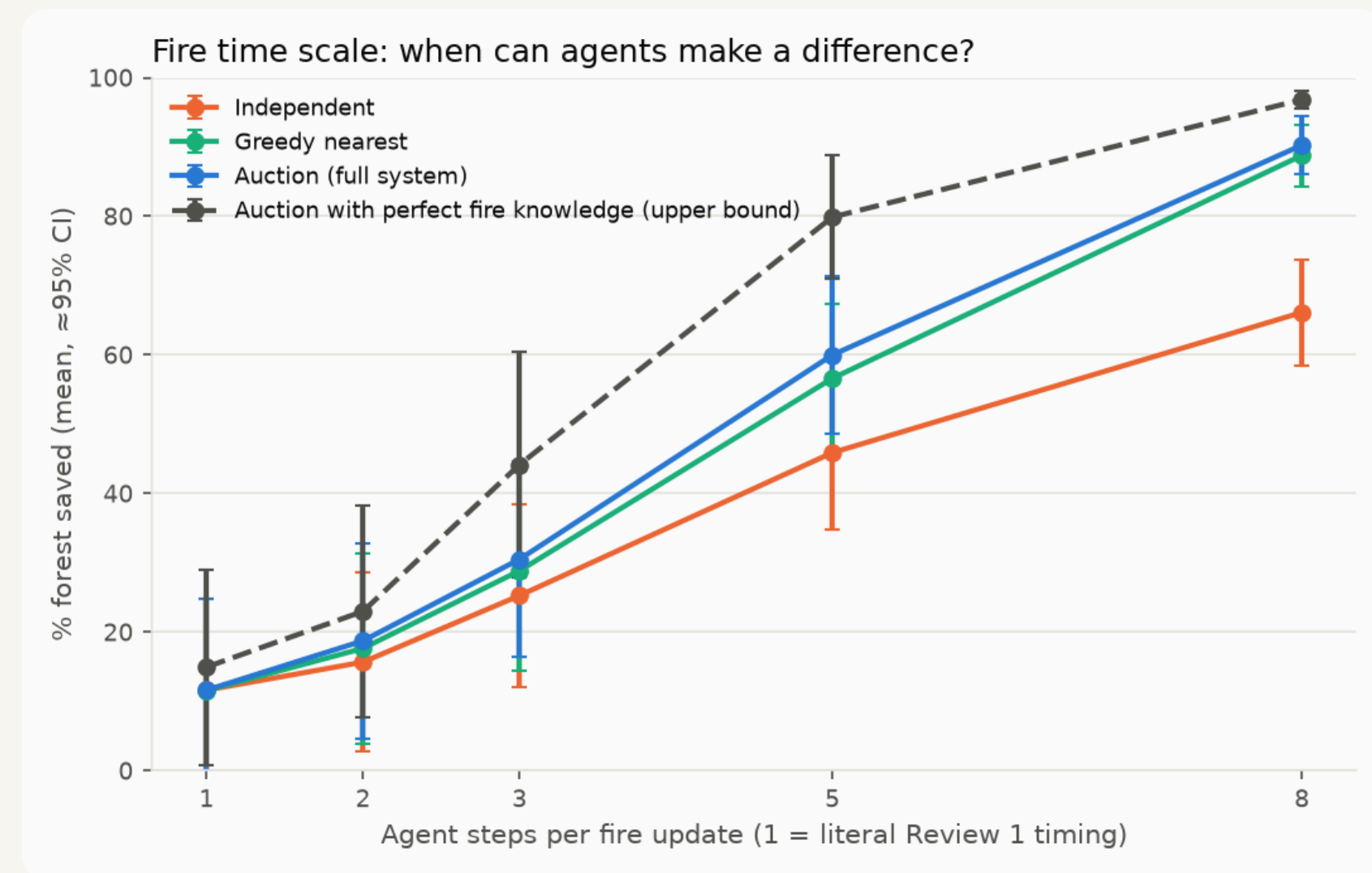
auction over greedy with three fires, the only significant difference

What we predicted, and what we found

Review 1 predicted	Verdict	Evidence
The full system saves the most	Confirmed	Best mean in 6 of 8; above independent in 7 of 8
Largest margin with strong wind (S2)	Confirmed	+13.5 points over independent
Largest margin with three fires (S3)	Partly	Only case where auction beats greedy, but 3 fires overwhelm 5 firefighters
The auction beats greedy in general	Not confirmed	Tied: one or two zones leave no real choice

Most of the gain comes from the shared map and coordinated targeting; the auction adds a measurable edge only when fires compete.

Every change from the plan has a study behind it



Agents act 5× per fire update. At the literal timing even perfect knowledge saves only 15%.

Utility targeting in a zone. $+8.1 \pm 4.2$ points over the plan's downwind-edge rule.

Safety first. Firefighters caught fell from 0.9 to 0.1 per run.

Fallback if the Coordinator fails. $+16.8 \pm 4.9$ points over no fallback.

CODE STRUCTURE AND SCALABILITY

0 to 30 agents with no code change

```
wildfire_squad/  
config.yaml  
model.py · belief_map.py · messages.py  
environment/ cells · forest · fire  
agents/ scout · firefighter · coordinator  
algorithms/ astar · auction · frontier ·  
zones  
strategies.py · scenarios.py  
app.py · run_experiments.py ·  
run_studies.py  
tests/ 114 tests
```

Pure algorithms: nothing in algorithms/ imports Mesa, so tests call them directly.

One config file: every parameter; a scenario is just an override.

Scales: 0 scouts, 1 to 30 firefighters, any ignitions, tested.

LIMITATIONS AND NEXT STEPS

What this model leaves out

Limitations

- 2D grid, no terrain slope
- Simplified fire spread
- Instant, perfect communication
- Agents act 5× per fire update

Next steps

- Message delays and losses
- A fully decentralised auction
- Real terrain and GIS maps

Thank you. Questions?